TUTORIAL NINE

Memory Organization (Part 2)

1. Considering a computer system with a 10-bit logical address. Translate the logical address 1000011011 to the corresponding physical address for the following two cases:

   a) Assuming paging is used, the page size is 128 bytes and the page table is as given in Figure Q4a.

   b) Assuming segmentation is used, the maximum segment size that the system can support is 256 bytes, and the segment table is as given in Figure Q4b.

0	01011
1	00010
2	00010
3	10101
4	01001
5	01100
6	11010
7	00110

Figure Q1a

0	00010000	010000000000
1	00001000	100000000000
2	00100000	010011010000
3	00011000	100001100000

Figure Q1b

2. Assume a computer system uses paging for memory allocation. Logical address consists of *x* bits for the page number and *y* bits for the offset.

   a) What is the page size?

   b) How many entries are there in the page table?

   c) Assuming each page table entry occupies 4 bytes, how many entries are there in the outer level page table if a two-level paging scheme is used? What is the format of logical address?

Virtual Memory (Part 1)

3. For each of the page replacement policies listed below, calculate the number of page faults encountered when referencing the following pages:

   0 1 6 0 3 4 0 1 0 3 4 6 3 4

   Assume the availability of 4 empty page frames.

   a) FIFO
   b) CLOCK
   c) LRU